

Module 5 Narrative – Artifact 3 enhancement

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CS-499 – Computer Science Capstone

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February 2, 2026

This artifact is the Weight Tracker App. The enhancement is to change the backend database store to Firebase Firestore. This will also add in the previous enhancements to authenticate and search and sort. The Weight Tracker app allows a user on android set a goal of weight loss or gain, add weights on various dates, then alert the user when the goal is reached. The app also has SMS alerts for when the goal is reached. This artifact was created in October 2025 as the main project in CS-360, mobile architecture and development.

This artifact encompasses several skills acquired in the computer science program. It produces a complete, usable, practical product. The enhancements make it into a full cloud-based application. It allows me to demonstrate my skills in application design, java programming, cloud computing, databases and android development.

In this enhancement, the artifact was improved by using Firebase Firestore as the backend database instead of a local database using SQLite. This allows access across multiple devices and keeps data stored securely in the cloud instead of on the user's device.

I have achieved the course outcomes that I planned to meet. This enhancement, in my assessment is complete for this phase. The backend database has been moved to Firebase Firestore. There is local caching of data and local sorting using the merge sort as was done in enhancement 2.

This enhancement fulfills and exceeds the planned outcome in module one. With this third enhancement my Android weight tracker app is now fully cloud based and can be used as a useful tool for health monitoring. The outcome coverage is what I had planned it to be, where I added the database functionality using the cloud database.

Enhancing and modifying the artifact was straightforward. Firebase Firestore was straightforward to use but I did switch to unique user IDs instead of email addresses as the key for the data. I also had some configuration to do in Firebase itself including security configuration. Testing revealed a few bugs including duplicate toast messages but in the end these were fixed and the app appears to be stable. Firebase Firestore is a cloud based NoSQL Document database. (Firebase Cloud Firestore documentation, n.d.) I had learned about this type of database in CS-340 Client/Server Development.

Works Cited

Google. (n.d.). *Firebase Cloud Firestore documentation*. Firebase.

<https://firebase.google.com/docs/firestore>